Let $\vec{m}_1 = (\sin\varphi, -\cos\varphi, 0)$ $\qquad \sigma_{m_1} = \sigma^x \sin\varphi - \sigma^y \cos\varphi$

$\vec{m}_2 = (\cos\varphi, \sin\varphi, 0)$ $\qquad \sigma_{m_2} = \sigma^x \cos\varphi + \sigma^y \sin\varphi$

$\vec{m}_3 = (0, 0, 1)$ $\qquad \sigma_{m_3} = \sigma^z$

$$|n\rangle = e^{i\frac{\theta}{2}\sigma_{m_1}} |s, s\rangle = \left(\cos\frac{\theta}{2} + i\,\sigma_{m_1} \sin\frac{\theta}{2}\right)|\uparrow\rangle$$

$$\langle n|\sigma_{m_1}|n\rangle = \langle\uparrow|\left(\cos\frac{\theta}{2} - i\sigma_{m_1}\sin\frac{\theta}{2}\right)\sigma_{m_1}\left(\cos\frac{\theta}{2} + i\sigma_{m_1}\sin\frac{\theta}{2}\right)|\uparrow\rangle$$

$$= \langle\uparrow|\cos^2\frac{\theta}{2}\,\sigma_{m_1} + \sin^2\frac{\theta}{2}\,\underbrace{\sigma_{m_1}\sigma_{m_1}}_{1}\sigma_{m_1} - i\sin\frac{\theta}{2}\cos\frac{\theta}{2}\,\underbrace{\sigma_{m_1}\sigma_{m_1}}_{1}$$

$$+ i\cos\frac{\theta}{2}\sin\frac{\theta}{2}\,\underbrace{\sigma_{m_1}\sigma_{m_1}}_{1}|\uparrow\rangle = \langle\uparrow|\sigma_{m_1}|\uparrow\rangle = 0$$

$$\langle n|\sigma_{m_2}|n\rangle = \langle\uparrow|\left(\cos\frac{\theta}{2} - i\sigma_{m_1}\sin\frac{\theta}{2}\right)\sigma_{m_2}\left(\cos\frac{\theta}{2} + i\sigma_{m_1}\sin\frac{\theta}{2}\right)|\uparrow\rangle$$

$$= \langle\uparrow|\cos^2\frac{\theta}{2}\,\sigma_{m_2} + \sin^2\frac{\theta}{2}\,\sigma_{m_1}\sigma_{m_2}\sigma_{m_1} - i\sin\frac{\theta}{2}\cos\frac{\theta}{2}\,\sigma_{m_1}\sigma_{m_2}$$

$$+ i\cos\frac{\theta}{2}\sin\frac{\theta}{2}\,\sigma_{m_2}\sigma_{m_1}|\uparrow\rangle$$

$$\sigma_{m_1}\sigma_{m_2}\sigma_{m_1} = (\vec{\sigma}\cdot\vec{m}_1)\underbrace{(\vec{\sigma}\cdot\vec{m}_2)(\vec{\sigma}\cdot\vec{m}_1)}_{\underbrace{\vec{m}_2\cdot\vec{m}_1}_{\emptyset} + i\,\vec{\sigma}\cdot\underbrace{\vec{m}_2\times\vec{m}_1}_{-\hat{z}}} = -i(\vec{\sigma}\cdot\vec{m}_1)(\vec{\sigma}\cdot\vec{m}_3) \Rightarrow$$

$$\vec{m}_2\times\vec{m}_1 = \begin{vmatrix} x & y & z \\ \cos\varphi & \sin\varphi & 0 \\ \sin\varphi & -\cos\varphi & 0 \end{vmatrix} = \hat{x}\cdot 0 - \hat{y}\cdot 0 + \hat{z}(-1) = -\hat{z}$$

$$\Rightarrow (-i)\left(\vec{m}_1\cdot\vec{m}_3 + i\,\vec{\sigma}\cdot\underbrace{\vec{m}_1\times\vec{m}_3}_{-\vec{m}_2}\right) = (-i)(-i\,\vec{\sigma}\cdot\vec{m}_2) = -\vec{\sigma}_{m_2}$$

$$\vec{m}_1\times\vec{m}_3 = \begin{vmatrix} x & y & z \\ \sin\varphi & -\cos\varphi & 0 \\ 0 & 0 & 1 \end{vmatrix} = \hat{x}(-\cos\varphi) - \hat{y}(\sin\varphi) = -\vec{m}_2$$

$$\sigma_{m_1}\sigma_{m_2} = (\vec{\sigma}\cdot\vec{m}_1)(\vec{\sigma}\cdot\vec{m}_2) = i\vec{\sigma}\,\underbrace{\vec{m}_1\times\vec{m}_2}_{\hat{z}=\vec{m}_3} = i\sigma_{m_3}$$

$$\sigma_{m_2}\cdot\sigma_{m_1} = i\vec{\sigma}\cdot\vec{m}_2\times\vec{m}_1 = -i\sigma_{m_3}$$

$$\Rightarrow \langle n|\sigma_{m_2}|n\rangle = \left(\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}\right)\langle\uparrow|\sigma_{m_2}|\uparrow\rangle$$

$$-i\sin\frac{\theta}{2}\cos\frac{\theta}{2}\ \langle\uparrow|\,i\sigma_{m_3}-(-i\sigma_{m_3})|\uparrow\rangle = \cos\theta\,\langle\uparrow|\sigma_{m_2}|\uparrow\rangle$$

$$+\sin\theta\,\langle\uparrow|\sigma_{m_3}|\uparrow\rangle = \sin\theta\,\langle\uparrow|\sigma^z|\uparrow\rangle = \sin\theta\cdot\frac{1}{2}$$

$$\langle n|\sigma_{m_3}|n\rangle = \langle\uparrow|\cos^2\frac{\theta}{2}\,\sigma_{m_3}+\sin^2\frac{\theta}{2}\,\sigma_{m_1}\sigma_{m_3}\sigma_{m_1}$$

$$-i\sin\frac{\theta}{2}\cos\frac{\theta}{2}\left(\sigma_{m_1}\sigma_{m_3}-\sigma_{m_3}\sigma_{m_1}\right)|\uparrow\rangle \longrightarrow$$

$$\sigma_{m_1}\sigma_{m_3} = \vec{m}_1\cdot\vec{m}_3 + i\vec{\sigma}\cdot\underbrace{\vec{m}_1\times\vec{m}_3}_{-\vec{m}_2} = -i\sigma_{m_2}$$

$$\sigma_{m_1}\sigma_{m_3}\sigma_{m_1} = -i\sigma_{m_2}\sigma_{m_1} = (-i)\left(\vec{m}_2\cdot\vec{m}_1 + i\vec{\sigma}\cdot\underbrace{\vec{m}_2\times\vec{m}_1}_{-\vec{m}_3}\right)$$

$$= (-i)^2\sigma_{m_3} = -\sigma_{m_3}$$

$$\rightarrow \langle\uparrow|\left(\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}\right)\sigma_{m_3} - i\sin\frac{\theta}{2}\cos\frac{\theta}{2}\,(-2i\sigma_{m_2})|\uparrow\rangle$$

$$= \langle\uparrow|\cos\theta\,\sigma_{m_3} - \sin\theta\,\sigma_{m_2}|\uparrow\rangle = \cos\theta\cdot\frac{1}{2}$$

Now, $\vec{\sigma} = \sigma^x \hat{x} + \sigma^y \hat{y} + \sigma^z \hat{z}$

$$= \sigma_{m_1} \hat{m}_1 + \sigma_{m_2} \hat{m}_2 + \sigma^z \hat{m}_3$$

where $\sigma_{m_1} = \hat{\sigma} \cdot \hat{m}_1$, $\sigma_{m_2} = \hat{\sigma} \cdot \hat{m}_2$

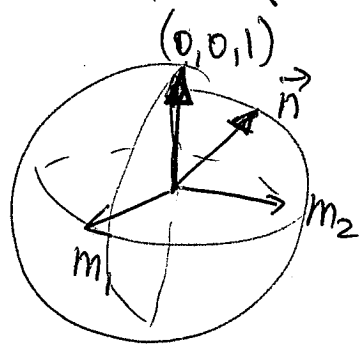
$$\Rightarrow \langle n | \hat{\sigma} | n \rangle = \hat{m}_1 \langle n | \sigma_{m_1} | n \rangle + \hat{m}_2 \langle n | \sigma_{m_2} | n \rangle$$

$$+ \hat{m}_3 \langle n | \sigma_{m_3} | n \rangle = \frac{1}{2} \hat{m}_2 \cdot \sin\theta + \frac{1}{2} \hat{m}_3 \cos\theta =$$

$$= \frac{1}{2}\left(\cos\varphi \sin\theta \, \hat{x} + \sin\theta \sin\varphi \, \hat{y} + \cos\theta \, \hat{z} \right)$$

$= S \vec{n}$ where

$$\vec{n} = \left(\sin\theta \cos\varphi, \ \sin\theta \sin\varphi, \ \cos\theta \right)$$



Thus $|\vec{n}\rangle = e^{i\frac{\theta}{2}\left(\sigma^x \sin\varphi - \sigma^y \cos\varphi\right)} |\uparrow\rangle$

represents state rotated by angle θ about axis $\vec{m}_1$.

$$S^x \sin\varphi - S^y \cos\varphi = \frac{S^+ + S^-}{2} \sin\varphi - \frac{S^+ - S^-}{2i} \cos\varphi =$$

$$= \frac{1}{2} S^+ \underbrace{(\sin\varphi + i\cos\varphi)}_{i e^{-i\varphi}} + \frac{1}{2} S^- \underbrace{(\sin\varphi - i\cos\varphi)}_{-i e^{i\varphi}} =$$

$$= \frac{i}{2}\left(S^+ e^{-i\varphi} - S^- e^{i\varphi} \right)$$

$$\Rightarrow |\vec{n}\rangle = \exp\left[-\frac{\theta}{2}\left(\hat{S}^+ e^{-i\varphi} - \hat{S}^- e^{i\varphi}\right)\right] |\tfrac{1}{2},\tfrac{1}{2}\rangle =$$

$$= \exp\left[z\,\hat{S}^+ - \bar{z}\,\hat{S}^-\right] |\tfrac{1}{2},\tfrac{1}{2}\rangle$$

where $\begin{cases} z = -\frac{\theta}{2} e^{-i\varphi} \\ \bar{z} = -\frac{\theta}{2} e^{i\varphi} \end{cases}$

We now can extend this to the state of arbitrary spin S:

$$|\vec{n}\rangle = e^{z\hat{S}^+ - \bar{z}\hat{S}^-} |S,S\rangle$$

Its main property: $\langle \vec{n} | \hat{\vec{S}} | \vec{n} \rangle = \vec{n}$

i.e. spin points along the direction of (arbitrary!) axis $\vec{n} = \sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$.

$$|n\rangle = e^{i\frac{\theta}{2}(\sigma_x \sin\varphi - \sigma_y \cos\varphi)} |\uparrow\rangle =$$

$$= \left[\cos\frac{\theta}{2} + i \sin\frac{\theta}{2} (\sigma_x \sin\varphi - \sigma_y \cos\varphi)\right] |\uparrow\rangle =$$

$$\sigma_x |\uparrow\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |\downarrow\rangle$$

$$\sigma_y |\uparrow\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i\, |\downarrow\rangle$$

$$= \cos\frac{\theta}{2} |\uparrow\rangle + i \sin\frac{\theta}{2} (\sin\varphi - i\cos\varphi) |\downarrow\rangle$$

$$= \cos\frac{\theta}{2} |\uparrow\rangle + e^{i\varphi} \sin\frac{\theta}{2} |\downarrow\rangle.$$

For spin $S > \frac{1}{2}$: $|S, S\rangle = \underbrace{|\uparrow\rangle \otimes |\uparrow\rangle \otimes |\uparrow\rangle}_{2S \text{ spins } \frac{1}{2}}$

Then $|\vec{n}\rangle = e^{i\theta(S_x \sin\varphi - S_y \cos\varphi)} |S, S\rangle$

becomes the tensor product of rotated states $\Rightarrow$ can derive everything for $S = \frac{1}{2}$ and then generalize to higher spins.

$$\langle n_2 | n_1\rangle = \left(\langle\uparrow| \cos\frac{\theta_2}{2} + \langle\downarrow| e^{-i\varphi_2} \sin\frac{\theta_2}{2}\right) \cdot$$

$$\cdot \left(\cos\frac{\theta_1}{2} |\uparrow\rangle + e^{i\varphi_1} \sin\frac{\theta_1}{2} |\downarrow\rangle\right) = \cos\frac{\theta_1}{2} \cos\frac{\theta_2}{2} \langle\uparrow|\uparrow\rangle$$

$$+ e^{i(\varphi_1 - \varphi_2)} \sin\frac{\theta_1}{2} \sin\frac{\theta_2}{2} \langle\downarrow|\downarrow\rangle =$$

$$= \cos\frac{\theta_1}{2} \cos\frac{\theta_2}{2} + e^{-i(\varphi_1 - \varphi_2)} \sin\frac{\theta_1}{2} \sin\frac{\theta_2}{2}.$$

$\Rightarrow$ for $S > \frac{1}{2}$: $\langle n_2 | n_1\rangle = \left[\cos\frac{\theta_1}{2} \cos\frac{\theta_2}{2} + e^{i(\varphi_1 - \varphi_2)} \sin\frac{\theta_1}{2} \sin\frac{\theta_2}{2}\right]^{2S}$

Completeness:

$$I = \frac{2s+1}{4\pi} \int d\vec{n} \; |n\rangle\langle n|$$

Check $s=\frac{1}{2}$: $\int d\vec{n} \; |n\rangle\langle n| = \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \cdot$

$$\cdot \left(\cos\frac{\theta}{2}|\uparrow\rangle + e^{i\varphi}\sin\frac{\theta}{2}|\downarrow\rangle\right)\left(\langle\uparrow|\cos\frac{\theta}{2} + \langle\downarrow|e^{-i\varphi}\sin\frac{\theta}{2}\right) =$$

$$= \int d\varphi \int d\theta \sin\theta \Big(\cos^2\frac{\theta}{2}|\uparrow\rangle\langle\uparrow| + \sin^2\frac{\theta}{2}|\downarrow\rangle\langle\downarrow|$$

$$+ e^{i\varphi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}|\downarrow\rangle\langle\uparrow| + e^{-i\varphi}\cos\frac{\theta}{2}\sin\frac{\theta}{2}|\uparrow\rangle\langle\downarrow|\Big)$$

But $\int_0^{2\pi} d\varphi \, e^{i\varphi} = \int_0^{2\pi} d\varphi (\cos\varphi + i\sin\varphi) = (\sin\varphi - i\cos\varphi)\Big|_0^{2\pi} = 0$

$$\int_0^{\pi} d\theta \sin\theta \cos^2\frac{\theta}{2} = \int_{-1}^{1} d(\cos\theta) \, \frac{1}{2}(1+\cos\theta) = \frac{1}{2}\int_{-1}^{1} dx(1+x) =$$

$$= \frac{1}{2}\left(2 + \frac{1}{2}x^2\Big|_{-1}^{1}\right) = 1 = \int_0^{\pi} d\theta \sin\theta \sin^2\frac{\theta}{2}$$

$$\Rightarrow \int d\vec{n} \; |n\rangle\langle n| = \left(|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|\right)\cdot 2\pi$$

$\Rightarrow$ unity $I = \frac{1}{2\pi}\int d\vec{n} \; |n\rangle\langle n|$ (for $s=1/2$) indeed.

Overlap integral for two adjacent time moments

$$\langle n(t+\delta)|n(t)\rangle = \cos\frac{\theta(t)}{2}\cos\frac{\theta(t+\delta)}{2} + e^{i(\varphi(t)-\varphi(t+\delta))}\cdot$$

$$\cdot \sin\frac{\theta(t)}{2}\sin\frac{\theta(t+\delta)}{2}$$

Now, $\theta(t+\delta) \simeq \theta(t) + \delta\theta'(t) = \theta + \delta\dot\theta$

$\varphi(t+\delta) \simeq \varphi(t) + \delta\varphi'(t) = \varphi + \delta\dot\varphi$

$$\langle n(t+\delta)|n(t)\rangle \simeq \cos\frac{\theta(t)}{2}\cos\left(\frac{\theta(t)}{2}+\frac{\delta}{2}\dot\theta\right) + e^{i(\varphi(t)-\varphi(t)-\delta\dot\varphi)}\cdot$$

$$\cdot \sin\frac{\theta(t)}{2}\sin\left(\frac{\theta(t)}{2}+\frac{\delta}{2}\dot\theta\right) \cong \cos\frac{\theta}{2}\left(\cos\frac{\theta}{2} - \frac{\delta}{2}\dot\theta\sin\frac{\theta}{2}\right)$$

$$+ e^{-i\delta\dot\varphi}\sin\frac{\theta}{2}\left(\sin\frac{\theta}{2} + \frac{\delta}{2}\dot\theta\cos\frac{\theta}{2}\right) =$$

$$= \cos^2\frac{\theta}{2} - \frac{\delta}{2}\dot\theta\underbrace{\left(\cos\frac{\theta}{2}\sin\frac{\theta}{2} - \sin\frac{\theta}{2}\cos\frac{\theta}{2}\,e^{-i\delta\dot\varphi}\right)}_{0+O(\delta)}$$

$$+ e^{-i\delta\dot\varphi}\sin^2\frac{\theta}{2} = \cos^2\frac{\theta}{2} + \left(1-i\delta\dot\varphi\right)\sin^2\frac{\theta}{2} =$$

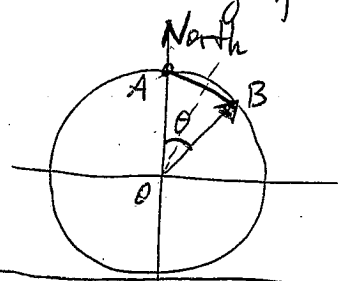
$$= 1 - i\delta\dot\varphi\,\frac{1}{2}(1-\cos\theta) = 1 - i\delta\left(\frac{1}{2}\right)(1-\cos\theta)\,\dot\varphi$$

$$\approx \exp -i\delta\left(\frac{1}{2}\right)(1-\cos\theta)\,\dot\varphi$$

$\Rightarrow$ for spin-S: $\langle n(t+\delta)|n(t)\rangle \approx \exp\left[-iS(1-\cos\theta)\dot\varphi\delta\right]$.

The meaning of the factor $(1-\cos\theta)\Delta\varphi$

Side View



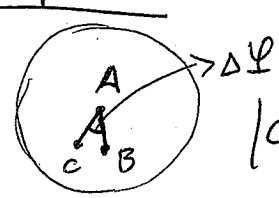
$\vec A = (0,0,1)$

$\vec B = (\sin\theta, 0, \cos\theta)$

$|\vec{AB}| = 2\sin\frac{\theta}{2}$

top view

$\vec B$ rotates into $\vec C$ in time δ.



$|\vec{CB}| = 2\sin\frac{\Delta\varphi}{2}\cdot|\vec{AB}|$

The area of triangle formed by unit vectors $\vec A, \vec B, \vec C$!

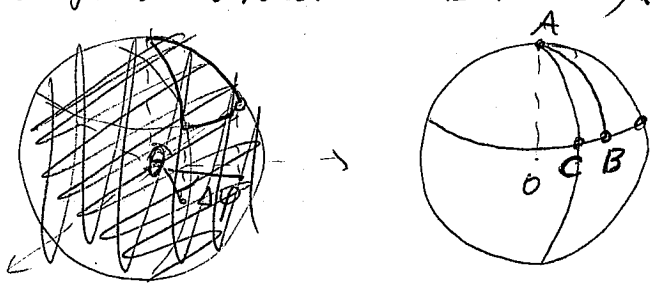
Area of $ABC = \frac{1}{2}|CB|\cdot\left(|\vec{AB}|\cos\frac{\Delta\varphi}{2}\right) =$

$\cong \frac{1}{2}\left(2\frac{\Delta\varphi}{2}\right)|\vec{AB}|^2 = 2\sin^2\frac{\theta}{2}\,\Delta\varphi$

The area formed by three unit vectors $\vec{A}$ (north pole), $\vec{B} = \vec{n}(t)$ and $\vec{C} = \vec{n}(t+\delta)$:

$$\Delta_{ABC} = 2\sin^2\frac{\theta}{2}\,\Delta\varphi = (1-\cos\theta)\,\Delta\varphi$$

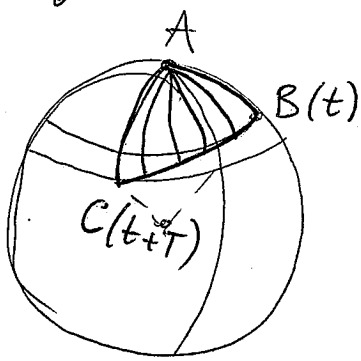
(to first order in $\Delta\varphi \ll 1$).

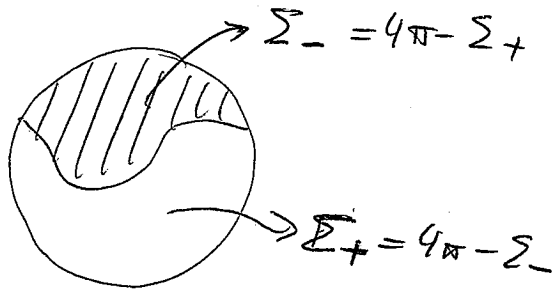


Imagine ~~the~~ electric (or magnetic) charge at the center of the sphere: then flux through the triangle ABC (area of which we calculated $2\sin^2\frac{\theta}{2}\Delta\varphi$) is equal to the flux through the spherical triangle ABC. [This is because of spherical symmetry: the flux is purely radial and passing through planar ABC passes through the spherical surface ABC]

Then $\int_0^T dt\,(1-\cos\theta(t))\,\dot{\varphi}(t)$ $\equiv$ area formed by $AB(t_0)C(t_0+T)$ (because it can be sliced into tiny segments)

$\equiv$ area of the cup Σ bounded by trajectory Γ parametrized by unit vector $\vec{n}(t)$





The area is not unique $\Rightarrow$ defined modulo 4π (total area of the unit sphere)